MUHAMMAD ZAHID	BT401 Genetic Resources & Conservation
MCQs	Fall 2020 (24 February)
Quiz 3	Final Term

1. Suitable light intensity range for most of the plant cultures is ___ µmol..... **10-1000**
2. Which of the following statement is correct about the Sanctuary?..... **All of the given**
3. Plant Genetic Resources Institute hosts the _____ lab/labs for plant genetic resources preservation.....**All given**
4. _____have created barriers to the migration of invertebrate species.....**Human activities**
5. Invertebrates accounts for _____ % of the animals on earth.....**95**
6. Forest trees are known for showing great _____ in their response to climate changes.....**Plasticity**
7. The act of moving plants or animals by human to different habitat to experience different environmental conditions is termed as _____..... **Assisted migration**
8. Before any plant or plant material is imported, an application for permit shall be submitted to the _____ .....**Entomologist**
9. _____organisms are described as the engineer of ecosystem.....**Earthworms**
10. The most important cause of extinction of animals and plants is_____.....**Habitat loss**
11. Some larger soil-dwelling invertebrates, such as earthworms, ants and termites, have been described as "ecosystem_____.....**Engineers**
12. Gene banks are also known as_____.....**Germplasm banks**
13. The natural protected area used for the conservation of genetic diversity is called _____.....**Gene sanctuaries**
14. The sum total of all the genes present in a crop and its related species constitutes its_____.....**Germplasm**
15. The lack of _____ may inhibit the ability of the population to respond to climate change.....**Genetic diversity**
16. Plants are totipotent because ____..... **A single cell could become a complete plant**

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17. Lyophilisation during DNA storage process helps to _____ the DNA……..**Preserve**

18. DNA banks are used for the storage of those species that cannot be conserved in _____ process of conservation……**Insitu**

19. _____protocol is used to access the genetic resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to the Convention on Biological Diversity……..**Nagoya protocol**

20. The Plant Genetic Resources Program (PGRP) is the main National Program that is functioning for overall management of _____ Plants……..**All given**

21. The quality of DNA extracted from plant specimens is dependent on____ **All of the given**

22. Which of the following statement truly describes the Biosphere Reserve?.....
protected areas to preserve genetic diversity in representative ecosystem

23. The usefulness of the specimen for PCR-based assays is greater than 4 years when stored at _____.....**-80^{0}C**

24. Germplasm can be quarantined in geographically and ecologically separated area to prevent the establishment of……..**Specific pathogen**

25. The Cartagena Protocol address handling, transport, packaging and identification of……..**All given**

26. The type of coiling in DNA is……**Right-handed**

27. A _____ is a large area of land where wild animals live safely or are hunted in a controlled way for sport……**Game reserve**

28. _____ensures the success of species in environments that are highly variable and subject to change……..**Genetic diversity**

29. Hybridisation can introduce the new genetic variation required by trees to adapt to……**Novel environments**

30. There are two main drawbacks of gene sanctuary one of which is that ……. **It cannot be easily maintained**

31. Field gene banks also called plant gene banks are areas of land in which _____ collections of growing plants are assembled……..**Germplasm**

32. Ex-suito conservation includes……**All given**

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33. Some larger soil-dwelling invertebrates, such as earthworms, ants and termites, help maintain the chemical fertility needed for……..**plant growth**

34. During tissue culture growth room temperature range from _____ requirements depending on species…….. **22 °C to 28 °C**

35. Which of the following statements is/are correct about Cartagena Protocol?.....**it is a biosafety protocol**

36. DNA of plants can be preserved by rapid drying of plant samples with the help of _____ material…….. **Silica gel**

37. The removal of the medium and transfer of cells from a previous culture into fresh growth medium, for further propagation of the cell line or cell strain is called……..**Subculturing**

38. The lack of genetic diversity may inhibit the ability of the population to respond to ____ change……..**climate**

39. Plant or plant material imported through the post shall be inspected by the _____ …….. **Plant Quarantine Officer**

40. Forests are important due to_____…….. **All given**

41. ___ factor/factors can affect the services of invertebrates to the environment…**All given**

42. How long a seed remains viable is known as its……..**Longevity**

43. The cartagena protocol on bio-safety was first established in……..**2000**

44. _____is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions……**Phenotypic plasticity**

45. Climate change related traits include……..**All given**

46. A situation where two (or more) species reciprocally affect each other's evolution is describes as……..**Co-evolution**

47. It has been estimated that at least ___ percent of world food production comes from crops that are dependent on insect pollination……..**35**

48. Almost 35% of world food production comes from crops that are dependent on……..**Insect pollination**

49. The first genomic libraries were cloned in……..**Plasmid**

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50. The ability of one genotype to produce more than one phenotype when exposed to different environmental conditions without genetic changes is called……**Phenotypic plasticity**

51. A formal written agreement 'contract, or other written instrument that establishes obligation……**Treaty**

52. In plant tissue culture, what is the term ORGANOGENESIS means?…… **Formation of root & shoot from callus culture**

53. Gene sanctuaries not only preserve the existing genetic diversity present in……**Allele and gene**

54. ____is the secure, long term storage of an individual's genetic material…….**DNA banking**

55. The process of transmission of characters from one generation to another is known as……..**heredity**

56. Plant and animal Quarantine regulation involves…… **All given**

57. Which type of animal or animal product can be transported through Pakistan Animal Quarantine Ordinance?……**All given**

58. What is the characteristic of the DNA that is used for the construction of the library…… **Plasmid and naked DNA**

59. ____ is not generally seen in biodiversity hotspots…..**Lesser interspecific competition**

60. Tree mortality in Ecosystem increases due to changes in ……**All given**

61. Forest help in driving……..**Water cycle**

62. Tree populations rely on interplaying mechanisms to respond to environmental change…… **All given**

63. Germplasm conservation under natural condition is called……**In-Situ conservation**

64. Animal and plant Quarantine Regulation is concerned with …….. **prevention of entry of diseased organisms**

65. Some insect use tree terpenes as precursor for their communication ____ to incorporate them into their own defense system…….. **pheromones**

66. _____ is an example of an Ex-situ conservation……**Seed bank**

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67. Pests and pathogens should be ranked by quarantine services according to the ____ they pose to crops……..**potential danger**

68. In 1984, gene sanctuaries for the conservation of wild relatives of coffee was setup in…….. **Ethiopia**

69. Seed orchards, clone banks and clonal archives are examples of ____ conservation……..**Static ex situ**

70. Gene sanctuaries are the type of _____.....**In-situ conservation**

71. A place or area where maximum variability of crop plants is observed is called ……**centers of diversity**

72. The adaptive capacity of ecosystem is more likely to dependent on……..**Invertebrates biodiversity**

73. Assisted migration may be undertaken…… **All given**

74. Germplasm conservation is important to conserve the _____ plants…….**endangered**

75. _____help to maintain soil structure and the availability of water throughout the soil profile……..**Earthworms**

76. Considering the better studied mammals and birds, 100% of the currently described species have been evaluated for their conservation statuses and, out of these, _____ out of 9,990 bird species are considered to be endangered……..**12%**

77. _____is defined as the number of species represented in a specific region, landscape or an ecological community…… **Species richness**

78. In _____ conservation of habitats rich in genetic diversity was recommended in the UN conference…….. **1972**

79. Most invertebrates are expected to change their geographical distribution…… **In response to climate change**

80. Forests majorly remove _____ from the atmosphere and produce oxygen……**carbon dioxide**

81. Candidate genes for drought tolerance include those involved in…… **All given**

82. What is Dimethyl sulfoxide used for?.....**Cryoprotectant**

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83. The capacity of ecosystem to adapt is diminished when biodiversity of _____ is lost…….**invertebrates**

84. What is Callus?..... **An unorganized actively dividing the mass of cells maintained in a culture**

85. Forest genetic resources or forest tree genetic resources are ___ of forest shrubs or tree species of actual or future ……**genetic material**

86. Ex-situ conservation also known as……..**static conservation**

87. ____ are needed to be able to cope with changing climatic conditions and increasing demand for food and retain the capacity to adapt to potential changes in the type of crops grown……**pollinator population**

88. Natural selection favour genotypes with……**All given**

89. Tissue culture conservation is an alternative to ……**vegetative propagation**

90. Perhaps the most neglected group of all in research, in farming practices, and in policies and strategies……**Invertebrates**

91. Diversity of forest genetic resources enable the species to adapt to…….. **All given**

92. Endemic species are ……..**species localized in a specific region**

93. Changes in climate also make it possible for ____ and diseases to invade new areas, destroying the forests there……. **pests**

94. Studies have shown that the presence of these animals can help to _____ the effects of drought on crop Production……. **Alleviate**

95. The sum total of all the genes present in a crop and its related species constitutes its…… **Germplasm**

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